

Three heads, two boards, no flying harness

A printable construction reference for a deliberately lo-fi Nano R4 looper. The panel board and core board mate through one keyed 2x15 header.

Scope and status

This guide freezes a **14HP prototype architecture**: three independently pitched loop heads, destructive overdub decay, Loop/One-shot playback, Record and Retrigger buttons, and matching CV inputs. It is an original Rainbow Demons design; it is not a clone of any commercial pedal.

Important: the electrical design has been cross-checked against published Nano R4 and 23LC1024 pin data, but the complete assembly and real-time firmware have **not** yet been bench-tested. Build and power it in stages.

What changed from the earlier build style

Item	Decision
Inter-board links	One direct, keyed 2x15 board-to-board header
Panel hardware	Pots, switches, buttons, LED and jacks mount on Panel Board
Processor/memory	Socketed Nano R4 and socketed 23LC1024 on Core Board
Serviceability	Four standoffs; boards unplug as one tested interface
Remaining wires	Only the standard Eurorack ribbon cable

Build order

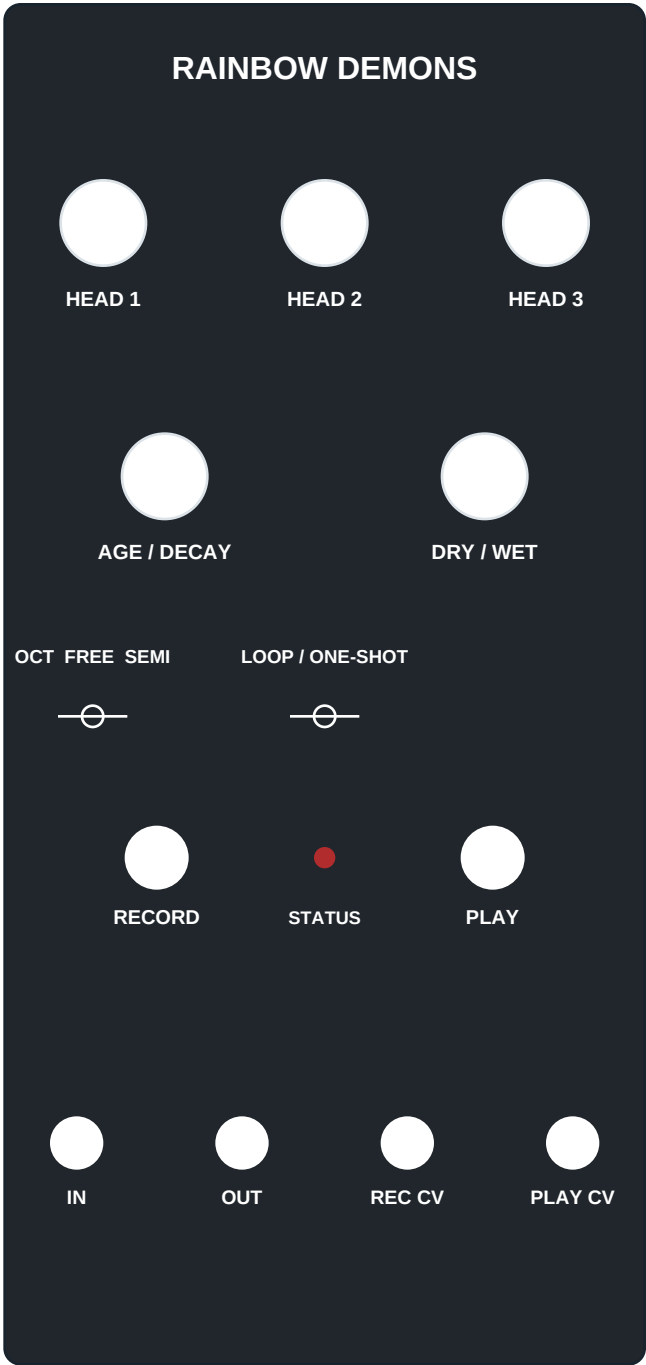
Stage	Stop and verify
1	Confirm panel parts and perfboard fit before drilling
2	Build and continuity-test the unpowered Panel Board
3	Build Core Board; fit IC sockets before the ICs
4	Power Panel Board alone; verify protected rails and 2.5V reference
5	Stack boards; verify header alignment and Nano supply
6	Flash smoke-test firmware, then audio firmware

Hard stop

Never connect USB and rack power during first bring-up. Remove the module from the rack before USB flashing, or open JP1 and unplug J1 so the analog board cannot be back-powered. Red stripe is -12V.

Panel and mechanical stack

Suggested 14HP ergonomics. Treat dimensions as a fit plan until your exact pots, jacks and switches are measured.



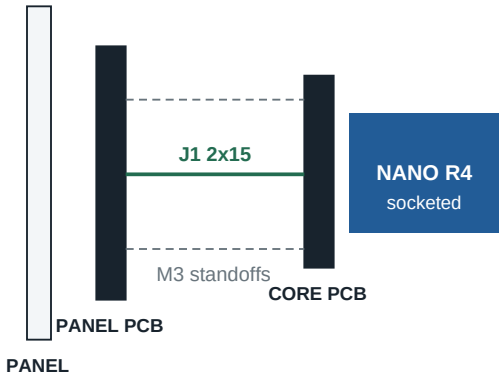
Panel envelope

Feature	Plan
Width	14HP = 70.8mm nominal
Height	3U panel = 128.5mm nominal
Panel PCB	Max suggested 64 x 108mm
Core PCB	Max suggested 64 x 90mm
Board gap	10-12mm female header + standoffs
Target depth	About 38-45mm behind panel

Before drilling

Print the panel sketch for arrangement only. Measure your actual bushing diameters, anti-rotation tabs, nut flats and body widths. Verify that the rear bodies clear J1 and all four standoffs.

Side view - front to rear



Header placement rule

Drill the four board mounting holes as a matched pair. Dry-stack J1 and the standoffs before soldering either header. Pin 1 must face the same upper-left corner on both boards.

Bill of materials

Choose through-hole parts where possible. Exact panel-component footprints must be confirmed before board drilling.

Qty	Part	Notes
1	Arduino Nano R4	ABX00142 / official Nano R4
1	23LC1024-I/P	PDIP-8, 1Mbit SPI SRAM
1	TL074CP	PDIP-14 quad JFET op amp
1	2x15 stack header pair	2.54mm, keyed by missing pin 30
2	1x15 female headers	Nano socket rails
1	DIP-8 socket	SRAM
1	DIP-14 socket	TL074
2	Isolated-pad perfboards	Panel 64x108; core 64x90 max
4	M3 standoffs	10-12mm plus screws/washers

Qty	Panel part	Notes
5	B10K pots	Head 1-3 center-detent preferred; Age and Mix linear
1	SPDT ON-OFF-ON	Quantize: octave / free / semitone
1	SPDT ON-ON	Loop / One-shot
2	Momentary NO buttons	Record and Play/Retrigger
1	3-lead bicolor LED	Common cathode + 2 x 1K resistors
4	3.5mm mono jacks	Audio in/out, Record CV, Play CV
1	2x5 shrouded header	Eurorack power; keyed notch
1	10-to-16 ribbon cable	Red stripe at -12V

Group	Required
Resistors	1K x9; 10K x12; 39K x1; 100K x4; 1M x3
Capacitors	100n ceramic x10; 1u film x1; 4.7u x1; 10u electrolytic x4
Semiconductors	2N3904 x2; 1N4148 x2; BAT54S x1; 1N5817 x2
Protection	F1 250mA + F2 100mA resettable fuses; JP1 header + shunt
Build	24-28AWG solid hookup wire, heat-shrink, labels, sockets

J1 - the complete board-to-board interface

One 2x15 header carries every rail, control and audio signal. Pin 30 is removed and its socket is blocked to prevent reversal.

PANEL BOARD - rear view



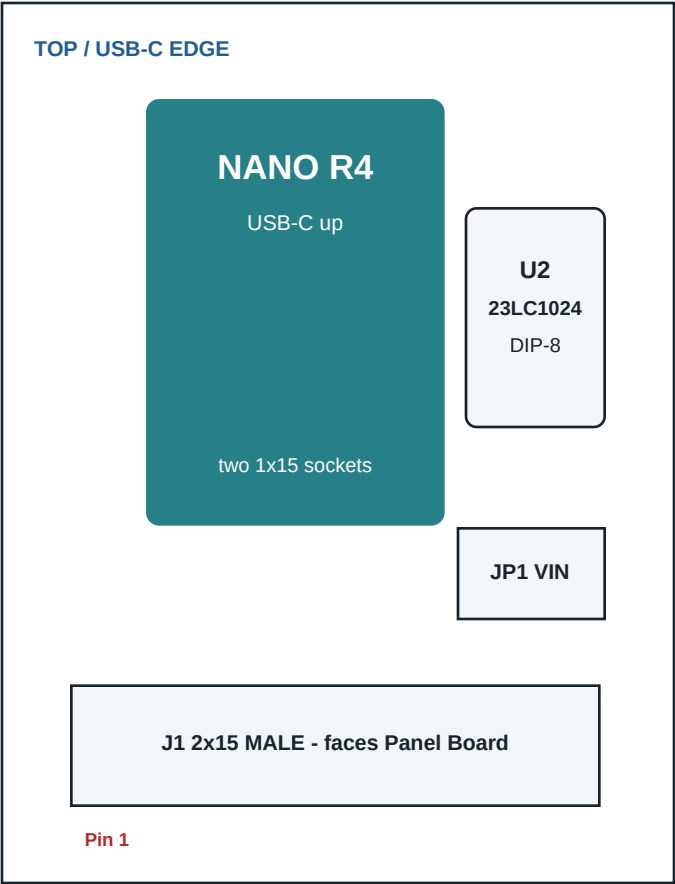
Pin	Net	Function
1	+12P	Protected +12V to JP1/Nano VIN
2	GND	Power return
3	+5N	Nano 5V out to Panel Board
4	GND	Analog return
5	AUDIO_ADC	Buffered, biased input to A1
6	GND	Input return
7	DAC_RAW	A0 DAC to output coupling cap
8	GND	Output return
9	HEAD1	Pot wiper to A2
10	GND	Control return
11	HEAD2	Pot wiper to A3
12	GND	Control return
13	HEAD3	Pot wiper to A4
14	GND	Control return
15	AGE	Pot wiper to A5
16	GND	Control return
17	MIX	Pot wiper to A6
18	GND	Control return
19	REC_BTN	Active-low to D2
20	PLAY_BTN	Active-low to D3
21	Q_OCT	Active-low to D4
22	Q_SEMI	Active-low to D5
23	PLAY_MODE	Active-low to D6
24	REC_CV	Active-low to D7
25	PLAY_CV	Active-low to D8
26	LED_R	D9 through 1K
27	LED_G	D0 through 1K
28	+5N	Second 5V contact
29	GND	Second ground contact
30	KEY	Pin removed; socket blocked

Assembly rule

Dry-fit the two boards, J1 and all four standoffs simultaneously. Tack only two diagonal header pins on each board, inspect for square alignment, then solder the rest. Mark pin 1 with permanent red paint.

Core Board - Nano R4 and loop memory

Component-side block plan and exact logic netlist. Use isolated pads; this is not a stripboard layout.



Nano pin allocation

Pin	Net	Use
A0	DAC_RAW	12-bit audio output
A1	AUDIO_ADC	Audio input
A2	HEAD1	Speed/direction
A3	HEAD2	Speed/direction
A4	HEAD3	Speed/direction
A5	AGE	Overdub retention
A6	MIX	Dry/wet crossfade
D2	REC_BTN	Falling edge / hold
D3	PLAY_BTN	Falling edge
D4	Q_OCT	Toggle throw
D5	Q_SEMI	Toggle throw
D6	PLAY_MODE	Loop / one-shot
D7	REC_CV	Conditioned trigger
D8	PLAY_CV	Conditioned trigger
D9	LED_R	Status red
D0	LED_G	Status green
D10	SRAM_CS	23LC1024 pin 1
D11	COPI	23LC1024 pin 5
D12	CIPO	23LC1024 pin 2
D13	SCK	23LC1024 pin 6

Core Board parts and wiring

U2 23LC1024: pin 1 CS to D10; pin 2 SO to D12; pin 3 SIO2 through 10K to +5N; pin 4 GND; pin 5 SI to D11; pin 6 SCK to D13; pin 7 HOLD through 10K to +5N; pin 8 +5N. Fit 100n directly across pins 8 and 4.

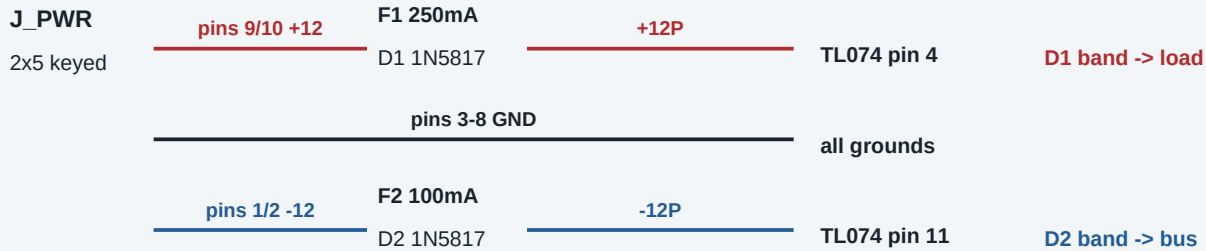
Analog controls: place 1K in series between J1 pins 9/11/13/15/17 and A2-A6. Place 100n from each Nano-side ADC node to GND. For audio, J1 pin 5 goes through 1K to A1; clamp the Nano side with BAT54S to GND and +5N.

Power: J1 pin 1 reaches VIN only through removable JP1. J1 pins 2/4/6/8/10/12/14/16/18/29 are one ground net. Nano +5V feeds J1 pins 3 and 28. Fit 100n plus 10u between +5N and GND near J1.

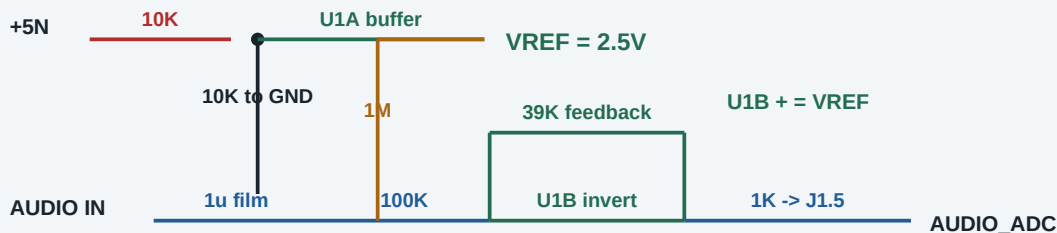
Panel Board - power and audio path

Build analog blocks first. TL074 uses protected +/-12V; the Nano uses protected +12V at VIN.

1. Eurorack power

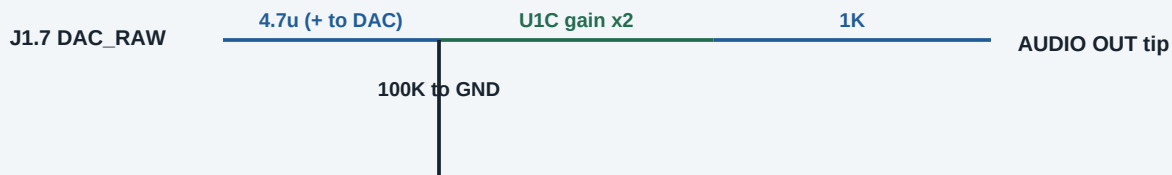


2. Midpoint reference and input



Input is AC-coupled and biased at VREF. VREF: 10u + 100n to GND.

3. DAC output



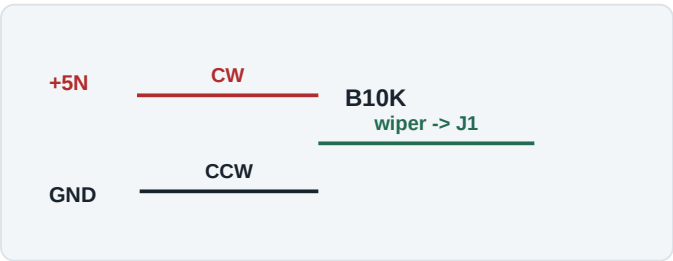
U1C: non-inverting; 10K feedback and 10K to GND. U1D spare: voltage follower at 0V.

Fit 100n at TL074 pins 4/11 and 10u from each protected rail to GND.

Panel Board - controls, triggers and LED

All panel parts terminate locally. Only named nets cross J1; there are no point-to-point wires between boards.

Pots - repeat five times



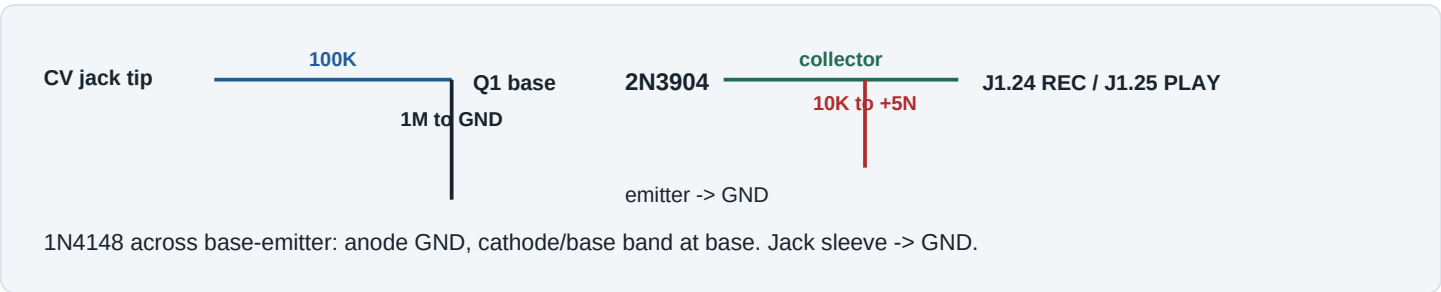
Head pots: J1.9, .11, .13
Age: J1.15
Mix: J1.17

Rear view warning: verify CW/wiper/CCW with a meter; pot lug numbering is not universal.

Manual controls - active low

Part	Net	Connection
Record NO	J1.19	Other terminal GND
Play NO	J1.20	Other terminal GND
Quant common	GND	Throws: J1.21 octave; J1.22 semitone
Mode common	GND	Throw: J1.23 one-shot; open = loop

CV inputs - build two identical channels



Status LED

Use a **3-lead common-cathode** red/green LED. J1.26 -> 1K -> red anode. J1.27 -> 1K -> green anode. Common cathode -> GND. Do not substitute a 2-lead inverse-parallel LED without changing the circuit and firmware.

Firmware contract and musical behavior

The hardware is pinned for a timer-driven 16kHz engine. Do not use blocking analogRead/analogWrite calls in the audio loop.

Parameter	Frozen requirement
Sample engine	16,000 samples/s, unsigned 8-bit companded storage
Loop capacity	131,072 bytes = 8.192 seconds maximum
Memory mode	23LC1024 sequential mode; SPI burst reads/writes
Input/output	A1 ADC sampled by timer; A0 12-bit DAC output
Unity zone	Center-detent head pots snap generously to exactly +1.000x
Head range	Target -3 to +3 octaves, with reverse through center mapping
Mix endpoints	Firmware snaps bottom 2% dry and top 2% fully wet

Control	Required behavior
Record	Empty: first press starts capture; next press closes loop. Existing loop: press toggles overdub. Hold about 2s clears and starts fresh.
Record CV	Each positive pulse performs the same edge action as the Record button.
Loop mode Play	First press starts/restarts continuous playback. Next press stops.
One-shot Play	Every press retriggers all three heads at loop start; playback stops at captured length.
Play CV	Same behavior as Play button; retrigger during a one-shot restarts immediately.
Age / Decay	Controls destructive overdub retention. CCW washes quickly; CW persists. Extreme CW may add a small, limited head-feedback path.
Quantize	Center is free. One throw rounds to octaves; the other rounds to semitone ratios.

Overdub equation - design intent

$$\text{new} = \text{softclip}(\text{old} \times \text{retention} + \text{input} \times \text{overdub_gain} + \text{heads_mix} \times \text{limited_regen}).$$
 Then apply a small low-pass loss, quantize to 8-bit and write back. The loop should fade and fray rather than remain clinically identical.

Assembly sequence

Complete each checkpoint before inserting the Nano R4, SRAM or TL074. A meter is part of the build, not an optional troubleshooting tool.

- 1 Mechanical dry fit**
Fit all panel hardware, both bare perfboards, J1 and four standoffs. Confirm rail clearance and USB-C access. Do not solder J1 until the stack is square.
- 2 Panel Board power**
Fit J_PWR, F1/F2, D1/D2, rail capacitors and JP1 feed. Mark red-stripe edge and diode bands. Leave ICs and Nano removed.
- 3 Panel Board analog**
Fit TL074 socket, VREF divider, input attenuator/buffer, DAC coupling/output stage and local decoupling.
- 4 Panel hardware**
Fit pots, toggles, buttons, LED, four jacks and both CV transistor stages. Continuity-map every actual jack lug.
- 5 Core Board**
Fit Nano sockets, SRAM socket, J1, ADC series resistors/capacitors, BAT54S clamp, pull-ups and decoupling.
- 6 Header continuity**
With boards separated, check every J1 pin against its destination and check adjacent pins for shorts. Confirm pin 30 is blocked.
- 7 Power alone**
Power Panel Board without Core Board. Verify +12P, -12P and no overheating. Fit TL074, then verify $V_{REF} = 2.5V \pm 0.1V$.
- 8 Stack and logic**
Power from a current-limited bench supply if available. Verify Nano VIN and +5N before fitting SRAM. Then fit SRAM and run smoke-test firmware.

Soldering discipline

Use insulated point-to-point links only on each perfboard. A crossing is not a junction. Clip exposed ends after inspection. Photograph both sides before stacking; label the photo with board revision and date.

Power-up and functional test

Use this order. If any reading fails, power down immediately and do not continue to the next section.

A. Unpowered continuity

- ☐ J_PWR red-stripe pins 1/2 reach only the protected -12P network through F2/D2.
- ☐ J_PWR pins 9/10 reach only +12P through F1/D1; neither rail is shorted to GND.
- ☐ J1 pin 1 reaches Nano VIN only when JP1 is fitted.
- ☐ J1 pin 3/28 is Nano +5N; it is not shorted to +12P, -12P or GND.
- ☐ J1 pin 30 is absent and the matching socket hole is blocked.

B. Panel Board powered alone

- ☐ +12P and -12P have correct polarity after the Schottky drops.
- ☐ TL074 pin 4 is positive, pin 11 is negative, and neither rail draws excessive current.
- ☐ VREF measures 2.5V +/-0.1V; AUDIO_ADC rests near 2.5V with no input.
- ☐ DAC output jack rests near 0V DC on the jack side of the coupling capacitor.

C. Stacked smoke test

- ☐ Nano VIN is approximately the protected +12V rail and Nano +5N is 4.75-5.25V.
- ☐ Each pot sweeps its ADC channel nearly 0-5V; center-detent head pots read near 2.5V.
- ☐ Each button/toggle input changes cleanly between HIGH and LOW in diagnostic firmware.
- ☐ Each CV input idles HIGH and produces one LOW pulse for a positive Eurorack trigger.
- ☐ SRAM walking-byte test passes across address 0x00000 through 0x1FFFF.

D. Audio test

- ☐ At Mix fully CCW, live input is clean and wet playback is absent.
- ☐ At Mix fully CW, live input is absent and captured playback remains.
- ☐ All three head pots at detent produce stable +1x traditional-looper playback.
- ☐ Loop/One-shot switch and both manual/CV retriggers follow page 8 exactly.

Quick start, faults and source record

Keep this page at the bench. Record measured values and firmware identity before calling the prototype complete.

Quick start

Action	Setting / result
Traditional loop	Heads 1-3 centered; Quantize FREE; Mode LOOP; Age 2 o'clock; Mix noon.
Capture	Press Record, play phrase, press Record again. Press Play if playback is stopped.
Overdub	Press Record to enter overdub; play; press again to close. Age sets wash rate.
One-shot	Select ONE-SHOT. Play button or CV restarts all heads from the beginning.
Multi-head	Move one head at a time away from detent; use quantize switch for repeatable ratios.

Fast fault isolation

Symptom	First checks
No power	Remove Nano. Recheck ribbon orientation, F1/F2, diode bands and JP1.
ADC stuck high/low	Separate boards; check J1 key, VREF, TL074 orientation and BAT54S.
Whine / digital hash	Shorten SPI and DAC paths; inspect decoupling and analog ground returns.
Loop corrupts	Run SRAM test; confirm HOLD/SIO2 pull-ups and sequential-mode command.
False CV triggers	Confirm transistor pinout, 1M base pulldown, reverse diode and firmware debounce.
No full wet/dry	Check Mix wiper sweep, endpoint snap, and that dry/wet gains reach exact zero.

Build record

Builder: _____ Date: _____ Board rev: _____

+12P: _____ V -12P: _____ V +5N: _____ V VREF: _____ V

Firmware file / commit: _____

SRAM: PASS / FAIL Audio: PASS / FAIL CV: PASS / FAIL

Primary references

Arduino Nano R4 hardware page and ABX00142 datasheet: <https://docs.arduino.cc/hardware/nano-r4>

Microchip 23LC1024 datasheet DS20005142: <https://ww1.microchip.com/downloads/en/DeviceDoc/20005142C.pdf>

Doepfer A-100 power-system pinout and red-stripe guidance: https://doepfer.de/a100_man/a100t_e.htm